Vidyavardhini's College of Engineering & Technology

Department of Computer Science Engineering (Data Science)

Academic Year: 2026-27

Academic Year: 2026-27

Subject: MDM- Wireless & Mobile Communication

Name:	Jay Chandrasen Talekar
Roll No & Branch:	50 / CSEDS
Class/Sem:	TE/V
Experiment No.:	08
Title:	To study and analyze path loss and signal attenuation in an indoor Wi-Fi environment by measuring Received Signal Strength Indicator (RSSI) variations across different physical obstacles (line-of-sight, wooden door, glass partition, concrete wall, plywood, and metal sheet) at a fixed distance.
Date of Performance:	
Date of Submission:	
Marks:	
Sign of Faculty:	

Aim

To study path loss in an indoor environment by measuring Wi-Fi signal strength (RSSI) under different obstacle conditions.

Theory

1. Path Loss

Path loss is the reduction in the power or strength of a wireless signal as it travels from the transmitter to the receiver.

In a wireless communication system, the signal becomes weaker as it travels through the environment due to distance, walls, obstacles, reflection, diffraction, and absorption.

Path loss is an important factor in determining the coverage and reliability of a wireless communication system.

2. RSSI (Received Signal Strength Indicator)

RSSI is a measure of the received signal strength at the receiver.

For Wi-Fi networks, RSSI is commonly represented in **dBm**.

RSSI values are generally negative. A value closer to zero indicates a stronger received signal.

For example:

- **−40 dBm** → Strong signal
- **−60 dBm** → Moderate signal
- **−75 dBm** → Weak signal

The exact signal-quality interpretation may vary depending on the device and environment.

3. Indoor Path Loss

In an indoor environment, wireless signals may pass through walls, doors, glass, furniture, and other obstacles.

These obstacles can absorb, reflect, or scatter the wireless signal, resulting in a reduction in received signal strength.

Therefore, indoor path loss depends not only on distance but also on the surrounding physical environment.

4. Effect of Obstacles

Different obstacles affect wireless signals differently.

Common indoor obstacles include:

- Walls
- Wooden doors
- Glass doors/windows
- Furniture
- Metal objects
- Multiple walls

A thicker or denser obstacle may cause greater signal attenuation.

5. Effect of Distance

As the distance between the Wi-Fi router and smartphone increases, the received signal generally becomes weaker.

Therefore, RSSI usually becomes more negative as the distance increases.

For example:

Short distance → stronger RSSI

Long distance → weaker RSSI

6. Signal Attenuation

Signal attenuation refers to the reduction in signal strength as the wireless signal travels through a medium or encounters an obstacle.

In an indoor environment, attenuation can occur due to:

- Walls and doors
- Furniture
- Human bodies
- Building materials
- Absorption
- Reflection
- Diffraction

Higher attenuation results in weaker received signal strength.

7. Relationship Between RSSI and Path Loss

RSSI can be used to observe changes in received signal strength under different indoor conditions.

When an obstacle is introduced between the transmitter and receiver:

Obstacle → Signal attenuation → Lower RSSI → Higher path loss

Therefore, comparing RSSI values under different conditions helps us understand the effect of obstacles on wireless signal propagation.

Procedure

Step 1: Select a Wi-Fi Router

Take a Wi-Fi router/access point and place it at a fixed location in the laboratory or classroom.

Keep the router switched ON throughout the experiment.

Step 2: Select a Smartphone

Take a smartphone having Wi-Fi capability.

Install and open a suitable **Wi-Fi Analyzer / Wi-Fi signal monitoring application** that can display the RSSI value of the connected Wi-Fi network.

Step 3: Connect the Smartphone to Wi-Fi

Connect the smartphone to the Wi-Fi network provided by the selected router.

Verify that the smartphone is successfully connected to the router.

Step 4: Fix the Measurement Distance

Select a suitable fixed distance between the router and smartphone, for example **5 m**.

Keep the approximate distance constant while comparing different obstacle conditions.

Step 5: Measure RSSI in Open Space

Place the smartphone such that there is no major obstacle between the router and smartphone.

Observe the RSSI value displayed by the Wi-Fi Analyzer application.

Record the measured RSSI value in the observation table.

Step 6: Introduce the First Obstacle

Place an obstacle between the router and smartphone, such as a **glass door/window**.

Keep the router, smartphone, and measurement distance approximately unchanged.

Step 7: Measure RSSI with the Obstacle

Observe the RSSI value with the obstacle present.

Record the RSSI value in the observation table.

Step 8: Test Different Obstacles

Repeat the experiment using different indoor obstacles, such as:

- Wooden door
- Glass door/window
- One wall
- Two walls
- Furniture, if applicable

Step 9: Record RSSI Values

For each obstacle condition, record the corresponding RSSI value.

If possible, take **2–3 readings** for each condition and calculate the average RSSI.

Step 10: Compare RSSI Values

Compare the RSSI values obtained under:

- Open-space condition
- Single-obstacle condition
- Different obstacle conditions
- Multiple-wall condition

Observe how the received signal strength changes.

Step 11: Perform Distance Testing

If required, repeat the experiment by changing the distance between the router and smartphone.

Suggested distances may be:

1 m → 3 m → 5 m → 7 m → 10 m

Record the RSSI value at each distance.

Step 12: Maintain Similar Experimental Conditions

During distance testing, try to keep:

- Router position fixed
- Smartphone orientation approximately fixed

- Same Wi-Fi network
- Similar environmental conditions

This helps in obtaining more consistent observations.

Step 13: Calculate Average RSSI

If multiple readings are taken at the same condition, calculate the average RSSI.

For example:

Average RSSI =
$(RSSI_1 + RSSI_2 + RSSI_3) / 3$

Step 14: Plot the Graph

Plot a graph of:

Obstacle Condition vs. RSSI

or, for distance testing:

Distance vs. RSSI

Observe the variation in signal strength.

Step 15: Analyze the Results

Compare the RSSI values obtained for different obstacles and distances.

Observe which conditions result in stronger or weaker received signals.

Step 16: Determine the Effect of Obstacles

Identify how walls, doors, glass, and other obstacles affect the received Wi-Fi signal.

Generally, an increase in obstacles causes greater attenuation and a weaker received signal.

Step 17: Record Observations and Conclusion

Record all experimental observations in the observation tables.

Write the conclusion based on the measured RSSI values and the effect of obstacles on wireless signal propagation.

OUTPUT

Observation Table – RSSI vs. Obstacles

Sr. No.	Obstacle / Condition	Distance (m)	RSSI (dBm)	Signal Observation
1	Open space	5	-44 dBm	Strong
2	Glass door/window	5	-48 dBm	Strong
3	Wooden door	5	-52 dBm	Moderate
4	One wall	5	-58 dBm	Moderate
5	Two walls	5	-66 dBm	Weak
6	Other obstacle	5	-55 dBm	Moderate

Observation Table – RSSI vs. Distance

Sr. No.	Distance (m)	RSSI (dBm)	Signal Observation
1	1	-38 dBm	Strong
2	3	-46 dBm	Strong
3	5	-52 dBm	Moderate
4	7	-59 dBm	Moderate
5	10	-67 dBm	Weak

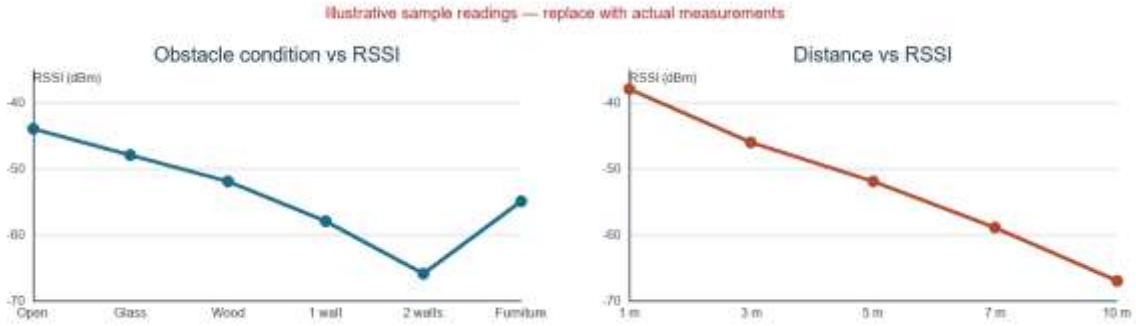
Graph

Graph 1: Obstacle Condition vs. RSSI

- X-axis: **Obstacle Condition**
- Y-axis: **RSSI (dBm)**

Graph 2: Distance vs. RSSI

- X-axis: **Distance (m)**
- Y-axis: **RSSI (dBm)**



Path Loss Analysis

Condition	RSSI Observation	Path Loss / Attenuation Observation
Open space	-44 dBm (strong)	Reference condition; least attenuation in this sample.
Glass obstacle	-48 dBm (strong)	4 dB weaker than open space; slight relative attenuation.
Wooden door	-52 dBm (moderate)	8 dB weaker than open space; moderate relative attenuation.
One wall	-58 dBm (moderate)	14 dB weaker than open space; greater relative attenuation.

Multiple walls	-66 dBm (weak)	22 dB weaker than open space; greatest relative attenuation in this sample.

General Observation

- Higher RSSI (closer to 0 dBm) indicates a stronger received signal.
- Lower RSSI (more negative value) indicates a weaker received signal.
- Obstacles generally reduce the received signal strength.
- Increasing the distance generally reduces the received signal strength.
- Different obstacles produce different amounts of signal attenuation.

Range / Signal Analysis

Condition	Signal Strength	Observation
Open space	Strong	Highest RSSI in this sample; fewest obstacles.
Single obstacle	Moderate	RSSI decreases when one obstacle blocks the path.
Multiple obstacles	Weak	Lowest RSSI in this sample; more obstacles cause greater attenuation.

Maximum observed attenuation	Two walls: 22 dB relative decrease from open space	Largest reduction in this dataset.

Using the readings, the signal changes from −44 dBm in open space to −66 dBm through two walls, a 22 dB relative decrease. At the same 5 m distance, the sample readings become progressively weaker from glass (−48 dBm) to a wooden door (−52 dBm), one wall (−58 dBm), and two walls (−66 dBm). In the distance sample, RSSI falls from −38 dBm at 1 m to −67 dBm at 10 m.

Conclusion:

The experiment demonstrates that indoor obstacles and greater separation can reduce received Wi-Fi signal strength. In the dataset, RSSI became more negative as obstacles increased and as distance grew, indicating greater attenuation and relative path loss.

Viva Questions

1. What is path loss in wireless communication?

Answer: Path loss is the reduction in signal power as a radio wave travels from a transmitter to a receiver.

2. What is RSSI?

Answer: RSSI is the Received Signal Strength Indicator, a receiver-side measure of the power level of a received signal.

3. Why is RSSI generally represented using negative dBm values?

Answer: The received power is below the 1 mW reference used for 0 dBm, so it is expressed as a negative dBm value. A value nearer 0 dBm is stronger.

4. What happens to RSSI when the distance between the transmitter and receiver increases?

Answer: RSSI generally becomes more negative as distance increases because the signal spreads and loses power while propagating.

5. How do walls and other obstacles affect wireless signals?

Answer: Walls and obstacles absorb, reflect, scatter, or diffract radio waves, usually reducing received signal strength.

6. What is signal attenuation?

Answer: Signal attenuation is the reduction in signal amplitude or power as it travels through a medium or encounters an obstacle.

7. What is the difference between RSSI and path loss?

Answer: RSSI is the received signal level measured at the receiver. Path loss is the reduction between transmitted and received power; estimating absolute path loss requires the transmit power and suitable system measurements.

8. Why should the router position be kept fixed during the experiment?

Answer: Keeping the router fixed controls one major variable so readings can be compared fairly across conditions.

9. Why should the smartphone orientation be kept approximately constant?

Answer: Phone orientation affects its antenna pattern and polarization, which can change the received signal level.

10. Which generally gives a stronger signal: open space or a condition with multiple walls?

Answer: Open space generally gives a stronger signal because fewer obstacles block or absorb the radio wave.

11. What is the relationship between distance and received signal strength?

Answer: As distance increases, received signal strength generally decreases and RSSI becomes more negative.

12. Name some common indoor obstacles that can cause signal attenuation.

Answer: Walls, wooden doors, glass, furniture, metal objects, and people can attenuate indoor Wi-Fi signals.

13. Why can two different obstacles produce different RSSI values?

Answer: Materials differ in thickness, density, water content, and electrical properties, so they absorb and reflect radio waves by different amounts.

14. What factors other than obstacles can affect Wi-Fi signal strength?

Answer: Router transmit power and channel, interference, distance, antenna direction, phone orientation, and building materials can all affect Wi-Fi signal strength.

15. What happens to wireless communication when path loss increases?

Answer: Higher path loss means less received power, which can reduce range, data rate, and link reliability or cause disconnections.